Variable laws: choose the independent variable

The best integration variable is the one already present in the acceleration law.

When acceleration is a function of time

Integrate the known function of time once for velocity and again for position. Each integration constant is fixed by a measured initial state.

$$v = v_0 + \int_{t_0}^{t} a(\tau)d\tau$$

Polynomial acceleration produces polynomial v and x; dimensions of every coefficient must match.

When acceleration is a function of velocity

Separate the differential equation. Equilibria where a(v)=0 are approached asymptotically in many dissipative models; do not divide through an equilibrium without checking it.

$$t - t_0 = \int_{v_0}^{v} \frac{dw}{a(w)}$$

WHY / A divergent time integral to an equilibrium signals asymptotic approach, not finite-time arrival.

When acceleration is a function of position

Apply the chain rule and integrate between measured endpoints. The result is a speed-energy relation even when no energy model has been introduced.

$$v^2(x) = v_0^2 + 2\int_{x_0}^{x} a(X)dX$$

WATCH / Choose the sign of v from direction; v² alone cannot tell whether motion is forward or returning.

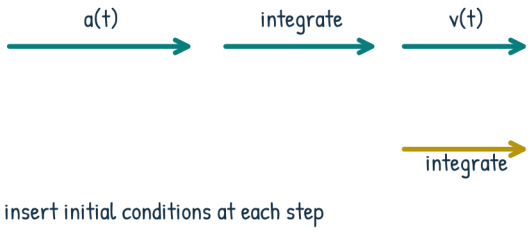
Turning points and forbidden regions

A candidate trajectory can occupy only positions where the computed v² is non-negative. A zero of v may be a turn, a stopping point, or an asymptote depending on acceleration and time integral.

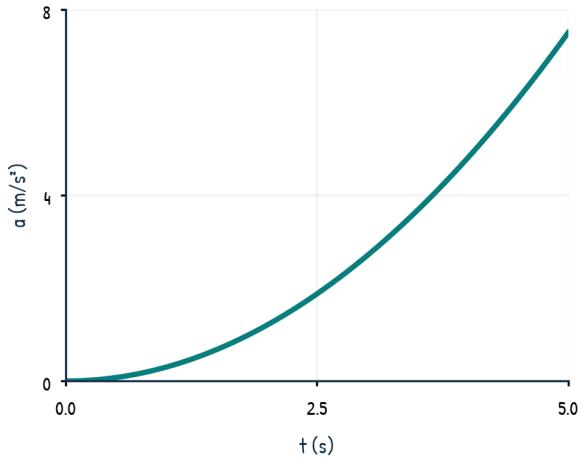
$$v^2(x) \geq 0 \quad \text{on the physical path}$$

Check both the sign branch and the time to reach each endpoint.

variable a: physical sketch



variable a: diagnostic graph



REMEMBER / a(t), a(v), and a(x) lead to different first-order integrals; preserve branches and initial data.

Linear drag and terminal velocity

A linear drag law is proportional to velocity relative to the surrounding medium.

Set the sign convention

For a body moving vertically in still air, choose downward positive. Weight is +mg and linear drag is -bv. The equation is first-order and dissipative.

$$m\frac{dv}{dt} = mg - bv, \qquad \tau = \frac{m}{b}$$

WATCH / Drag opposes velocity relative to air; if the body rises, drag points downward too.

Solve from rest

The equilibrium speed is the terminal speed. Subtract it from v to obtain an exponential decay equation for the speed difference. For times much shorter than the time constant, expansion gives velocity near gt and fall distance near half gt squared.

$$v_T = \frac{mg}{b} = g\tau, \qquad v(t) = v_T(1 - e^{-t/\tau})$$

At t=0, v=0; as t grows, v approaches vT from below.

Position and early-time limit

Integrating velocity gives the fall distance from rest. At short times the exponential expands to recover near-free-fall motion.

$$y - y_0 = v_T[t - \tau(1 - e^{-t/\tau})]$$

WHY / For t much smaller than tau, v is approximately gt; at long times position becomes nearly linear in time.

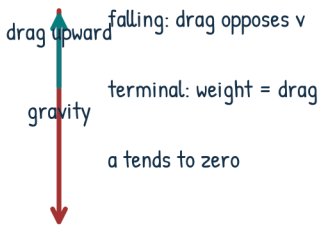
Terminal speed is an equilibrium

At terminal speed, net acceleration is zero: drag balances weight. The velocity does not become exactly terminal at finite time in this model.

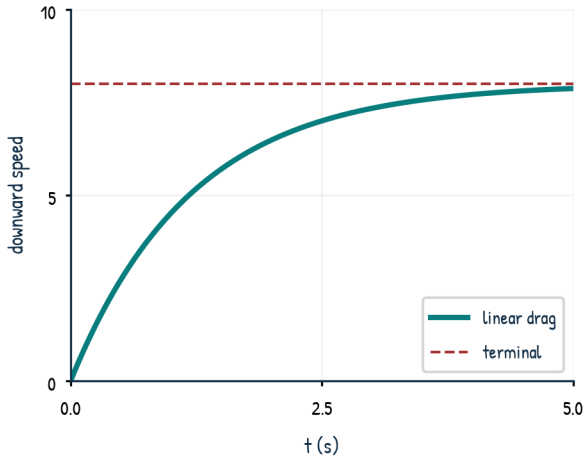
$$\lim_{t\to\infty} a = 0, \qquad \lim_{t\to\infty} v = v_T$$

As b tends to zero, tau and vT grow and the finite-time solution tends to free fall.

drag1d: physical sketch



drag1d: diagnostic graph



REMEMBER / For downward-positive fall: v(t)=vT(1-e^(-t/tau)) from rest, with tau=m/b and vT=gtau.

Quadratic drag and terminal fall

At larger Reynolds number, drag magnitude is often modelled proportional to speed squared.

State the quadratic law

For downward velocity v>=0, write drag magnitude kv² upward. The signed equation balances weight against drag; at v=0, drag vanishes and the initial acceleration is g.

$$m\frac{dv}{dt} = mg - kv^2$$

WATCH / For motion in either direction, the vector law is proportional to -v|v|, not always -v².

Terminal speed

At equilibrium, mg=kvT². Dividing the differential equation by mg exposes a separable logistic-like form. Starting from rest, the terminal speed is approached asymptotically, not reached at finite time.

$$v_T = \sqrt{\frac{mg}{k}}, \qquad \frac{dv}{dt} = g(1 - v^2/v_T^2)$$

The terminal speed increases with mass and decreases with the square root of drag coefficient.

Integrate from rest

Separate variables and integrate. The hyperbolic tangent satisfies the initial condition and approaches one at large argument.

$$v(t) = v_T \tanh\left(\frac{gt}{v_T}\right)$$

WHY / At early time tanh zaboutz, giving vaboutgt; at late time v approaches vT.

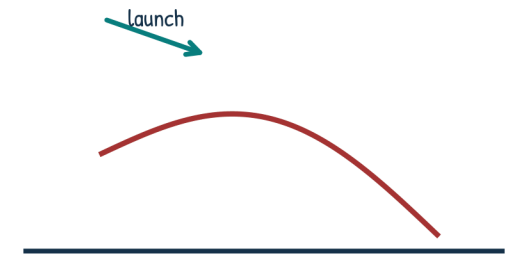
Do not confuse drag regimes

Linear and quadratic drag are idealisations whose use depends on Reynolds number, shape and flow. The coefficient k is not universal across objects or speeds.

$$Re = \frac{\rho_{air} v L}{\eta}$$

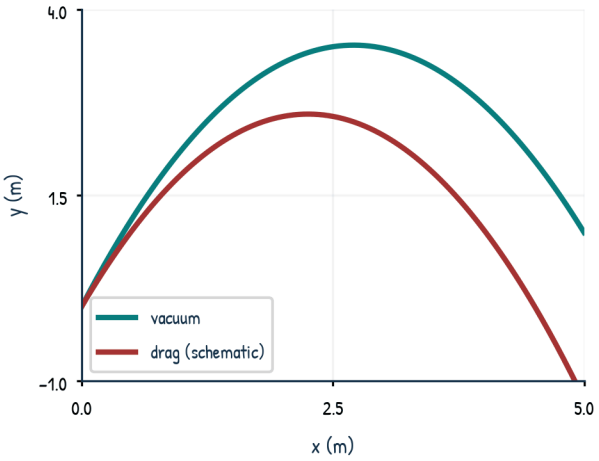
A measured terminal speed can determine a drag parameter only after the force law and projected area are specified.

drag2d: physical sketch



drag breaks vacuum symmetry

drag2d: diagnostic graph



REMEMBER / For downward-positive fall from rest: v=vT tanh(gt/vT), vT=sqrt(mg/k).

Stopping, reaction time and scaling

A human response delay and a braking force are distinct stages.

Reaction phase

During a fixed reaction interval tr, speed is approximately unchanged. The vehicle travels forward before braking starts; this distance is already covered before the assumed braking acceleration acts.

$$d_{\text{reaction}} = v_0 t_r$$

At 20 m/s and 0.5 s, the reaction distance is 10 m.

Braking phase

For constant deceleration magnitude b, use the no-time equation from initial speed v0 to rest. This is braking distance only.

$$d_{\text{brake}} = \frac{v_0^2}{2b}, \qquad t_b = \frac{v_0}{b}$$

WATCH / The assumed b is constant and independent of speed; real tire-road limits may change the model.

Combine and compare

Total distance is the sum of phases. Doubling speed doubles reaction distance but quadruples braking distance when the reaction time and b are fixed.

$$d_{\text{total}} = v_0 t_r + \frac{v_0^2}{2b}$$

WHY / At high speed, braking distance dominates; at low speed, reaction distance can be a substantial fraction.

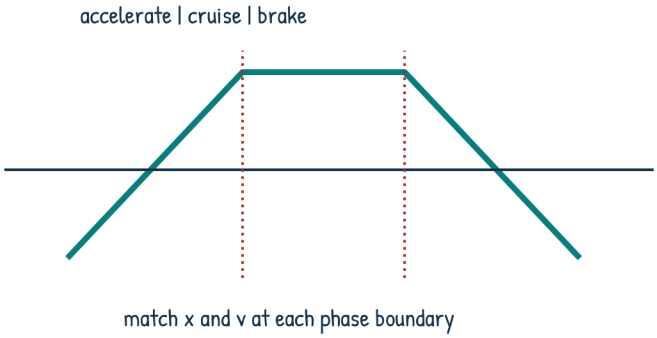
Check a full example

For v0=20 m/s, tr=0.5 s and b=5 m/s², reaction distance is 10 m and braking distance 40 m; the total is 50 m. Substitution in $v^2 = u^2 - 2bd = 0$ checks the final speed at the end of braking.

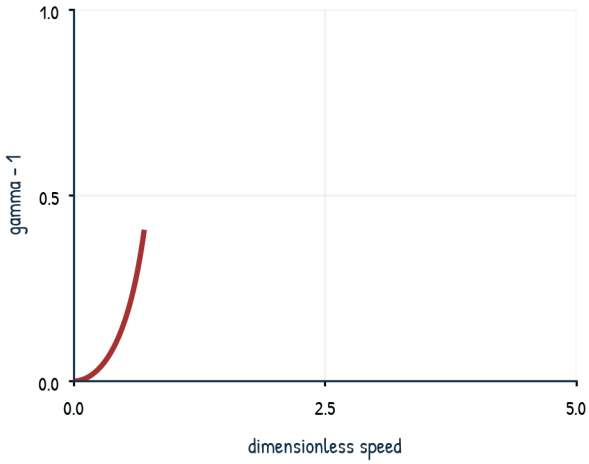
$$d_{\text{total}} = 10 + 40 = 50 \text{ m}$$

Changing the assumed reaction time changes only the first term.

piecewise: physical sketch



model: diagnostic graph



REMEMBER / Reaction distance scales as v; constant-deceleration braking distance scales as v².

Projectile with linear drag

For isotropic linear drag, horizontal and vertical ODEs decouple, but the landing event still matters.

Equations in still air

With up positive, drag force is -b v as a vector. Let tau=m/b. Horizontal velocity decays exponentially; vertical velocity also has a terminal offset.

$$\dot{v}_x = -v_x/\tau, \quad \dot{v}_y = -g - v_y/\tau$$

WATCH / This model is isotropic and linear in velocity; it is not the standard quadratic high-speed drag law.

Integrate the components

For launch components v0x and v0y from the origin, integrate each velocity with x(0)=y(0)=0. Horizontal position approaches tauv0x as tto∞ even though landing generally occurs earlier.

$$x(t) = \tau v_{0x}(1 - e^{-t/\tau})$$

$$y(t) = \tau(v_{0y} + g\tau)(1 - e^{-t/\tau}) - g\tau t$$

Find the actual landing time

Landing at the launch elevation requires the positive nonzero root of y(T)=0. In general this transcendental equation is solved numerically; use the root consistent with the descending branch.

$$y(T) = 0, \quad v_y(T) < 0$$

WATCH / The integral of vx to infinite time is not the projectile range; the projectile has already landed.

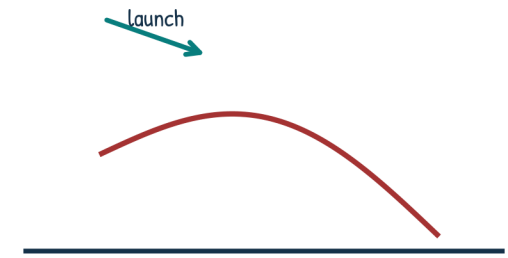
Then evaluate range and limits

The horizontal range is x(T). As b tends to zero, the equations tend to the vacuum trajectory and the landing time tends to 2v0y/g.

$$R = x(T), \qquad b \to 0 \Rightarrow R \to \frac{v_0^2 \sin 2\theta}{g}$$

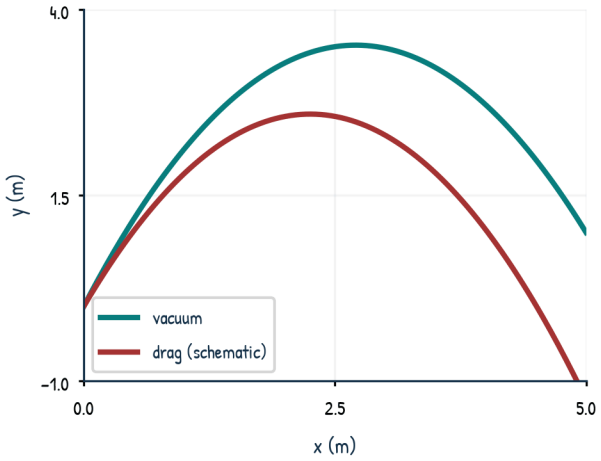
A correct range calculation must use one common finite T for both coordinates.

linear-drag projectile: physical sketch



drag breaks vacuum symmetry

linear-drag projectile: diagnostic graph



REMEMBER / Solve x(t), y(t); find the positive ground-contact time from y(T)=0, then evaluate x(T).

Constant horizontal deceleration

A deliberately simple headwind model adds a constant horizontal acceleration.

Separate the components

Assume a constant horizontal acceleration -aw throughout flight and no vertical drag. Then vertical motion is unchanged, so equal-height flight time is still set by v0y and g.

$$x = v_{0x}t - \tfrac{1}{2}a_w t^2, \quad y = v_{0y}t - \tfrac{1}{2}gt^2$$

WATCH / A real aerodynamic headwind generally gives a speed-dependent force, not a constant deceleration.

Range correction

At the equal-height landing time T=2v0y/g, evaluate x(T). The change from vacuum range is half aw T squared; aw is a positive deceleration magnitude opposing the initial horizontal motion.

$$R = v_{0x}T - \tfrac{1}{2}a_w T^2 = R_0 - \tfrac{1}{2}a_w T^2$$

As aw tends to zero, recover the vacuum range.

Forward landing condition

A positive horizontal acceleration is not implied; if deceleration is strong enough, horizontal velocity can reverse before landing and x(T) may be zero or negative.

$$v_x(T) = v_{0x} - a_w T$$

WHY / Distinguish signed horizontal displacement from path length when the projectile reverses horizontally.

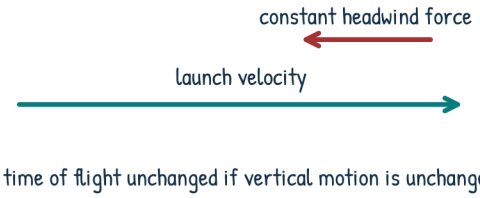
Angle dependence

For fixed launch speed, a steeper angle increases flight time and therefore increases the correction aw T²/2 in this model. The optimum launch angle is no longer generally 45 degrees.

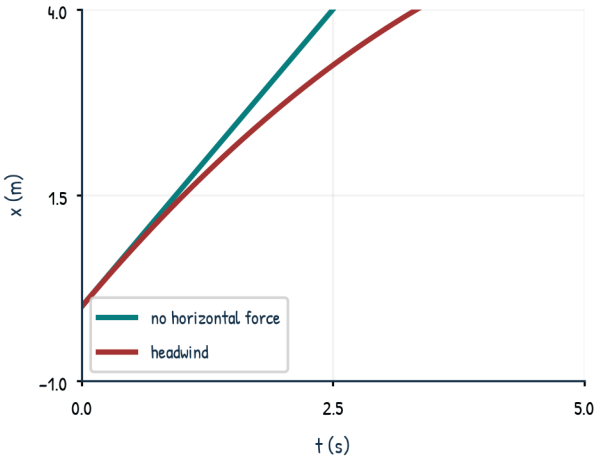
$$\Delta R = -\frac{2a_w v_0^2 \sin^2\theta}{g^2}$$

This expression has dimensions of length and vanishes at zero flight time.

headwind: physical sketch



headwind: diagnostic graph



REMEMBER / Vertical motion fixes T; horizontal displacement includes the acceleration term.

Interception with gravity: solve at a common time

A moving target and projectile must occupy the same position at one shared time.

Write both position laws

Let the target begin at r0 and move at constant velocity vt. The projectile starts at the origin with unknown launch vector v0 and acceleration qvec.

$$\mathbf{r}_p = \mathbf{v}_0 t + \tfrac{1}{2}\mathbf{g}t^2, \qquad \mathbf{r}_T = \mathbf{r}_0 + \mathbf{v}_T t$$

WATCH / Do not treat the target as stationary in the projectile frame unless the frame transformation is applied to both velocity and acceleration.

Impose interception

Set positions equal and solve for the required launch vector as a function of candidate time. Fixed launch speed imposes one scalar equation.

$$\mathbf{v}_0(t) = \tfrac{\mathbf{r}_0}{t} + \mathbf{v}_T - \tfrac{1}{2}\mathbf{g}t$$

WHY / Each positive solution time gives a candidate launch direction; reject candidates that hit the ground or an obstacle first.

Fixed-speed condition

For prescribed v0 magnitude, solve the squared magnitude equation. Gravity makes this generally a quartic in t; there may be zero, one, or several physical solutions.

$$|\mathbf{v}_0(t)|^2 = v_0^2$$

A no-gravity interception reduces to a quadratic condition in 1/t after choosing the frame.

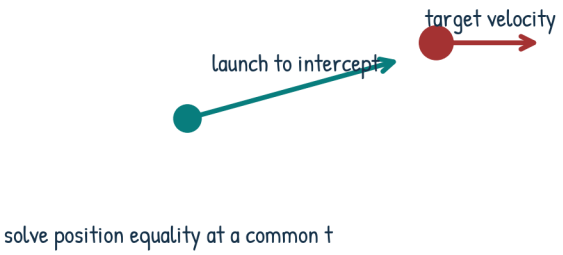
Physical root audit

Check t>0, the initial and target positions, velocity direction, and whether the target is actually reachable under the model. Root counting is part of the solution.

$$\mathbf{r}_p(T) = \mathbf{r}_T(T), \qquad T > 0$$

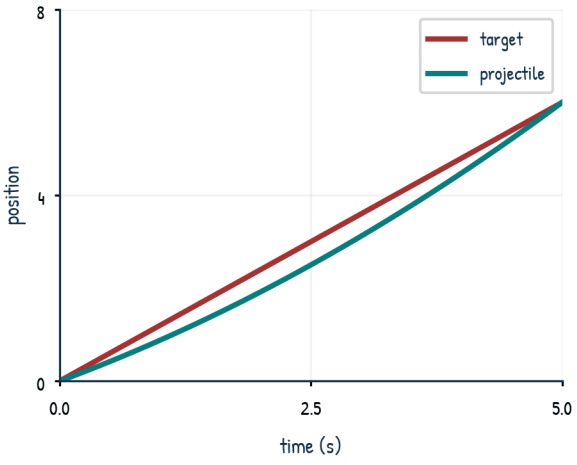
WATCH / "Aim directly at the target's current position" is generally wrong when the target moves or gravity acts.

intercept: physical sketch



solve position equality at a common t

intercept: diagnostic graph



REMEMBER / For fixed launch speed, impose the target position equation on the projectile trajectory and solve for positive t.

Coriolis deflection: first-order estimate

In a rotating Earth frame, horizontal motion acquires a small sideways acceleration.

Rotating-frame acceleration

In a frame rotating with angular velocity Omega, the Coriolis term is -2 Omega cross v. It is perpendicular to velocity and changes direction, not speed, to first order.

$$\mathbf{a}_C = -2\boldsymbol{\Omega} \times \mathbf{v}$$

WATCH / This is an apparent acceleration in the rotating frame, not a new interaction force in an inertial frame.

Local northward motion

Resolve Earth rotation into local north and vertical components. For horizontal northward motion at latitude lambda, the horizontal Coriolis acceleration points east in the Northern Hemisphere.

$$a_E \approx 2\Omega v_N \sin\lambda$$

At the equator, this horizontal component vanishes for purely northward motion.

Estimate sideways displacement

If the northward speed and flight time T are nearly constant, treat the small eastward acceleration as constant over the interval.

$$\delta_E \approx \tfrac{1}{2} a_E T^2 = \Omega v_N T^2 \sin\lambda$$

WHY / This first-order estimate neglects feedback of the sideways velocity into Coriolis acceleration.

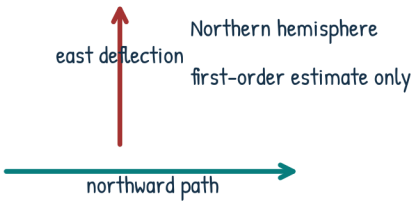
Numerical scale

At latitude 45 degrees, with vN=500 m/s and T=20 s, the estimate is about 10.3 m east. Use actual flight time from vertical motion, not range divided by speed unless the geometry warrants it.

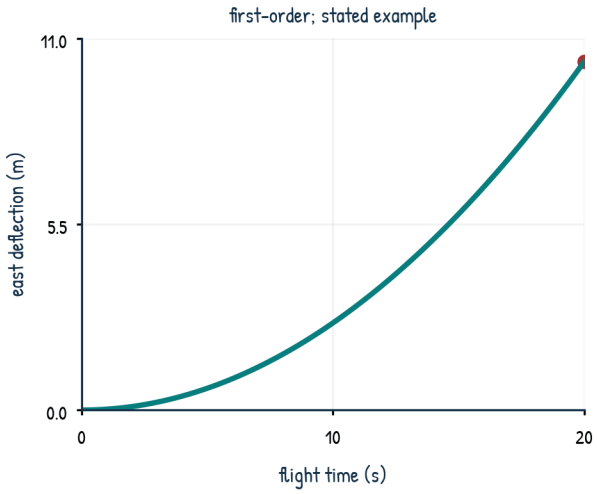
$$\Omega \approx 7.29 \times 10^{-5}\ \mathrm{s}^{-1}, \quad \delta_E \approx 10.3\ \mathrm{m}$$

The deflection scales linearly with Omega, vN, and sinlambda, and quadratically with time.

coriolis: physical sketch



coriolis: diagnostic graph



REMEMBER / For northward speed v at latitude lambda, eastward Coriolis acceleration is about 2Omegav sinlambda in the Northern Hemisphere.

Planar pursuit: specify geometry before solving

A pursuit problem is a coupled ODE, not a universal power-law curve.

State the setup explicitly

Let the target start at the origin and move along +x at constant speed vt. A pursuer starts at (0,d), has constant speed vp, and always aims at the target's current position.

$$\mathbf{r}_T = (v_T t, 0), \qquad \mathbf{r}_P(0) = (0, d)$$

WATCH / Changing the initial geometry changes the path; no single pursuit curve applies to every setup.

Write the pursuit equations

Let dx and dy be the pursuer coordinates, and D the instantaneous distance to target. Its velocity is vp times the unit vector toward the target.

$$D = \sqrt{(v_T t - x)^2 + y^2}$$

$$\dot{x} = v_P \frac{v_T t - x}{D}, \quad \dot{y} = -v_P \frac{y}{D}$$

Use relative range for capture

For target-minus-pursuer separation vector r, its radial rate is at most vt-vp. If vp>vt, the separation decreases at a rate bounded away from zero, so capture occurs in finite time.

$$\dot{D}_{\text{sep}} \le v_T - v_P < 0$$

The bound gives capture time no greater than initial separation divided by vp-vt.

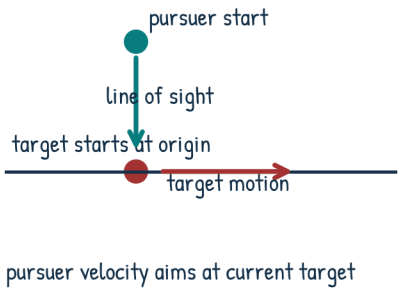
Solve the path with the stated data

The ODE generally requires numerical integration for the trajectory and capture time. At each step update target position and point the pursuer along the current line of sight.

$$\frac{d\mathbf{r}_P}{dt} = v_P \frac{\mathbf{r}_T - \mathbf{r}_P}{|\mathbf{r}_T - \mathbf{r}_P|}$$

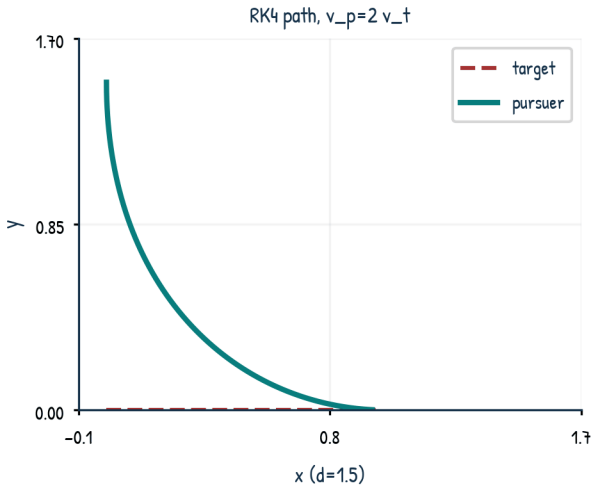
WHY / A claimed closed form must satisfy the ODE and the initial conditions; checking only dimensions is not enough.

pursuit: physical sketch



pursuer velocity aims at current target

pursuit: diagnostic graph



REMEMBER / With target at (vt t,0) and pursuer initially at (0,d), derive the line-of-sight ODE before seeking a path.

Model limits: Newtonian, vacuum and smooth paths

A correct formula includes the regime in which it is meaningful.

Newtonian speed boundary

At high speed, relativistic momentum is gammamv rather than mv. The fractional correction is approximately $v^2/(2c^2)$ when v is small compared with c.

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \approx 1 + \frac{v^2}{2c^2}$$

A 1 percent momentum correction begins near vabout0.14c.

Separate identity from model

The chain rule identity a=v dv/dx is kinematic. Constant acceleration, uniform gravity, vacuum projectile motion, and a chosen drag law are additional model assumptions.

$$\text{identity} \neq \text{model assumption}$$

WHY / A derivation can be algebraically correct and still use the wrong physical model.

Check low and high limits

Remove a small effect by taking its parameter to zero: drag coefficient to zero, wind speed to zero, or Earth rotation Omega to zero. The result should reduce to the simpler model.

$$b \to 0: \text{ vacuum}; \quad \Omega \to 0: \text{ no Coriolis term}$$

WATCH / Do not extrapolate a low-speed drag law or constant-acceleration approximation beyond its regime.

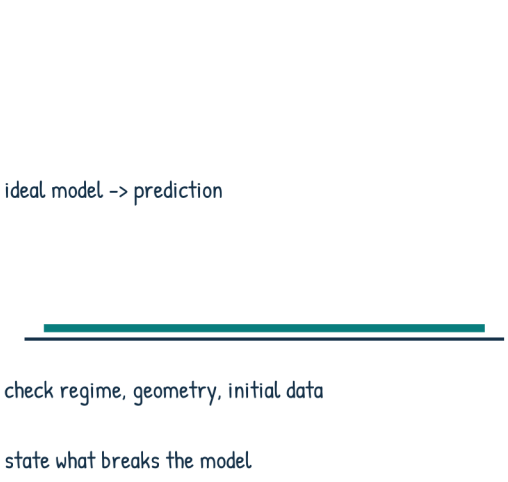
State domain and event

A trajectory formula describes motion only until the first physical constraint or contact event. Roots need positive time, and a path after impact is not part of the same free-flight phase.

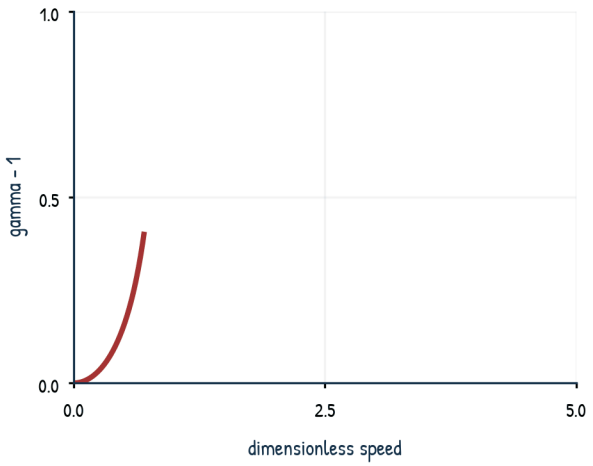
$$t \geq 0, \qquad \text{stop the model at first contact}$$

Before reporting a range, identify which surface is hit and at what time.

model: physical sketch



model: diagnostic graph



REMEMBER / Use Newtonian kinematics for v much less than c; distinguish model assumptions from algebraic identities.